

3 Month Intensive MERN Stack Internship.

Duration: 3 Months (12 Weeks)

Mode: Practical + Project-Based

Outcome: Full-Stack Developer – Job-Ready

Month 1 – Web Fundamentals & JavaScript Core

Week 1: Web & HTML Basics

- How the web works: Browser, Server, HTTP basics
- Static vs Dynamic websites
- HTML5 document structure
- Headings, paragraphs, links, images, lists, tables
- Semantic HTML elements (header, footer, nav, section, article)
- Forms: input types, labels, placeholders
- Client-side validation using HTML attributes
- Reusable layout sections (header, footer, sidebar)
- Accessibility basics
- Structuring pages for readability and SEO

Week 2: CSS Basics

- CSS syntax: inline, internal, external
- Box model (margin, padding, border)
- Fonts, typography scaling, color systems
- Backgrounds, borders, shadows
- Positioning: static, relative, absolute, fixed
- Flexbox layout: direction, alignment, spacing
- CSS Grid: row and column layout, responsive grids
- Media queries and responsive design
- Mobile-first design principles
- Cross-browser compatibility
- CSS variables and reusable component styles

Week 3: JavaScript Fundamentals

- Variables, data types, type conversions
- Operators, conditions, loops
- Functions, arrow functions, modular JS coding
- Arrays and objects
- String and array methods
- DOM manipulation (querySelector, getElementById, innerHTML)
- Event handling (click, submit, change, keyboard events)
- Basic form validation using JS
- Console debugging

Week 4: Advanced JavaScript

- ES6+ features: let/const, template literals, destructuring, spread/rest operators, modules
- Functions: callbacks, higher-order functions
- Arrays: map, filter, reduce
- Objects: key-value iteration, object destructuring
- JSON: parse and stringify
- Fetch API for HTTP requests
- Promises and async/await
- Handling errors with try/catch

Month 2 – React & Frontend Development

Week 5: React Fundamentals

- React project setup with Create-React-App
- Folder structure for scalable projects
- JSX syntax and usage
- Functional components
- Props for data passing
- State management with useState
- Controlled forms
- Conditional rendering
- Rendering lists and keys
- Event handling in React

Week 6: Advanced React

- useEffect hook for side-effects
- Component reusability patterns
- Lifting state up
- Form validation in React
- Handling API responses in frontend
- Error and loading state management
- Styling React components (CSS modules, inline, styled-components)
- React Router for page navigation
- Component folder organization

Week 7: Node.js & Express Basics

- Node.js runtime and event loop
- Creating a basic server
- Node modules and npm packages
- Express setup
- Routing (GET, POST, PUT, DELETE)
- Request and response objects
- Middleware functions
- Structuring backend code (controllers, routes)
- Error handling

Week 8: MongoDB & Backend Integration

- Database fundamentals (SQL vs NoSQL)
- MongoDB collections and documents
- CRUD operations: Create, Read, Update, Delete
- Mongoose schemas and models
- Database validation
- Connecting Node.js backend with MongoDB
- Handling asynchronous operations
- Querying database using filters and sorting

Month 3 – Advanced MERN, Authentication, Projects

Week 9: Authentication & Security

- User signup and login systems
- Password hashing with bcrypt
- JWT token generation and verification
- Middleware for protected routes
- Role-based access (Admin/User)
- Securing APIs
- Session management
- Error handling in authentication

Week 10: Minor Full-Stack Project

- CRUD-based application (e.g., Student Management System)
- React frontend
- Node.js + Express backend
- MongoDB database
- Authentication basics
- GitHub repository management
- Proper commit history.

Week 11: Major Full-Stack Project Part 1

- Project selection: Job Portal / E-Commerce / Task Manager
- Complete frontend UI development
- Implement routing and state management
- Backend API design for full application
- Database schema for production-like environment
- Authentication and role-based features
- Complex CRUD workflows

Week 12: Major Full-Stack Project Part 2

- Full frontend-backend integration
- Handling API responses and errors
- Component reuse and optimization
- Folder structure for maintainable code
- GitHub repository with professional commit patterns
- Project documentation
- Testing functionality
- Preparing project for interview explanation

Professional Skills & Practices Covered

- Git & GitHub workflow
- Clean code principles
- Modular and scalable project structure
- Debugging frontend & backend
- Handling asynchronous operations
- API integration
- Error handling & edge-case management
- Reusable component and logic patterns

Final Outcome After 3 Months

- Build complete MERN stack applications independently
- Frontend UI with React, state management, and routing
- Backend API with Node.js and Express
- Database design with MongoDB and Mongoose
- Authentication and role-based access control
- End-to-end CRUD workflows
- GitHub portfolio with structured commits
- Prepared for Full-Stack Developer job interviews